

Tobechukwu David Uka

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PROFESSIONAL SUMMARY

Results-driven senior software engineer (6+ years) building secure, scalable banking and payments systems. I design microservices and APIs for onboarding/KYC, accounts and balances, transactions and ledgers, lending workflows, and internal tools. Strong in data integrity, auditability, observability, and end-to-end delivery.

TECHNOLOGIES

Back-End & Cloud	Node.js, Python, FastAPI, Django, Flask, Pydantic, pytest, Hypothesis, mypy, black, isort, flake8, REST, GraphQL, GitHub Actions, Docker, Discord, Typescript, Kafka, RabbitMQ, Azure, AWS, SQS, SQL Server, SQL Server, PostgreSQL
AI & Machine Learning	LLMs; SFT; RL; prompt design; AI agents (tool/API calling); Retrieval-Augmented Generation (RAG); Embeddings and Vector Search (pgvector, Pinecone); Chunking and Indexing; Reranking; Function/Tool Calling; Benchmark Design; Schema-validated outputs (Zod/JSON Schema)
Front-End	ReactJS, NextJS, Angular, JavaScript, jQuery TypeScript, HTML5, CSS3
DevOps & Agile	CI/CD pipelines, Git, Code Reviews, Scrum, Automated Testing
Leadership	Team Building, Mentorship, Architectural Design, Stakeholder Engagement

EXPERIENCE

Lead Product Engineer | Sterling Bank | Lagos, Nigeria | May 2024 – Present

- Owned three loan products: Edubanc, EdPay for Teachers, and Edubanc Middleware.
- Designed end-to-end loan flows: sign-up and Know Your Customer (KYC), wallet and loan creation, loan disbursement, repayments, penalties, settlements, and reconciliation.
- Built the Edubanc Middleware as a multitenant platform in with tenant-aware routing, per-tenant Role-Based Access Control (RBAC) and throttling, configuration isolation, and data partitioning with encryption; automated tenant onboarding and provisioning.
- Implemented double-entry ledgers, clear audit trails, and safe webhooks with retries for wallets and direct debits.
- Integrated third-party wallets and direct-debit systems with idempotent webhooks and retries, improving transaction efficiency by about 40%.
- Shipped admin tools and live dashboards for operations, risk, and finance (approvals, settlement views, exception queues, and reporting exports).
- Used Artificial Intelligence (AI) to read and flag issues in bank statements and documents, speeding up review and reconciliation.
- Delivered the EdPay platform and reusable integration guides for schools and partners, boosting user engagement and platform growth by about 30%.
- Applied secure coding practices (RBAC, input validation, secrets management), enforced access controls and audit logs, and tuned performance for high-volume financial transactions.

Senior Software Engineer (POD LEAD) | Turing (Contract) | California, US | Nov 2024 – January 2025

- Led Python quality assurance for AI training programs (LLM fine-tuning/evaluation), owning QA gates for Python coding tasks, datasets, and code reviews.
- Ran regular spot checks on Python submissions for correctness, complexity, edge cases, and tests; delivered clear, actionable feedback and escalated issues when needed.
- Authored and maintained coding guidelines, style guides, rubrics, trackers, FAQs, and “honeypot” tasks to raise signal and catch low-quality work.
- Built reference implementations and fixtures in Python using FastAPI/Django/Flask; designed REST/GraphQL stubs and wrote unit/property tests (pytest, Hypothesis).
- Enforced consistency with mypy, black, isort, flake8, and pre-commit; added CI gates (GitHub Actions) for linting, types, tests, and schema checks.
- Coordinated trainers/QAs on Discord: posted updates, answered questions, triaged blockers, and kept the community engaged and aligned with changes.
- Monitored activation and output quality; reached out to inactive contributors to identify blockers and re-engage them with targeted guidance.
- Scheduled and delivered onboarding/training sessions; produced playbooks and short videos so new contributors could reach quality quickly.
- Built and iterated internal tools and eval dashboards used by 20,000+ engineers to track quality metrics, error types, and regression trends.
- Drove continuous improvement via root-cause analyses; tagged errors (logic gap, hallucination, I/O) and fed insights back into prompts, tasks, and guidelines.
- Partnered with product/data to scope new Python task types across algorithms, APIs, and real-world cases; maintained taxonomy, difficulty, and coverage.

Software Engineer (Full Stack) | Heirs Holdings | Lagos, Nigeria | Oct 2023 – Nov 2024

- Optimized Converge ERP workflows for 40,000 employees across a \$6.9B portfolio, improving processing efficiency by 50% through backend performance tuning, SQL query optimization, caching, and workflow refactors.
- Designed and delivered ERP workflow modules as services with well-defined REST APIs; produced conceptual and detailed technical designs aligned to enterprise standards.
- Built event-driven processing for workflow and notification pipelines using RabbitMQ to improve reliability, ordering, and async job throughput for high-volume internal operations.
- Worked with Service Management and Operations to plan releases, monitor system health, and manage incidents and post-release fixes; introduced runbooks and measurable operational checks.
- Developed the TEF Connect platform supporting 50,000+ entrepreneur registrations and pitch sessions, using scalable backend services and secure API integrations.
- Streamlined 13th-month allowance processing by implementing strict validation rules, consistent state handling, and automated reconciliation, increasing accuracy to 99.99% and reducing manual effort by 60%.
- Implemented a gamification module backed by rule-based scoring services and persistent storage (SQL/NoSQL where appropriate), increasing idea generation by 30%.
- Automated leave allowance processing with backend workflows and scheduled jobs, reducing admin workload by 75% and improving turnaround time.
- Containerized key services using Docker (and Kubernetes where applicable) to standardize deployments and reduce environment drift; improved release consistency and rollback safety.
- Documented the legacy codebase and integration points, reducing new-hire onboarding time by 50% and improving maintainability.

Solution Engineer (Full Stack) | DipoleDIAMOND | Lagos, Nigeria | Jan 2023 – Nov 2023

- Designed and shipped bank-integrated backend services and chat/payment workflows, producing solution designs aligned to enterprise architecture and security standards; implemented core services in Python with well-structured

REST APIs.

- Led a team of 5 to build chatbot solutions for tier-one and tier-two banks and 4 corporate clients, delivering secure integrations and reliable production rollouts.
- Integrated vendor and bank APIs, enforcing authentication, authorization, and audit logging; worked closely with Security and compliance teams to meet regulatory requirements.
- Built asynchronous processing for payment and notification workflows using Kafka where appropriate, improving resiliency and reducing failures during peak usage.
- Worked across SQL and NoSQL databases (MongoDB) for conversation state, event logs, and operational reporting, optimizing performance with indexing and caching.
- Coordinated test plans and supervised UAT; supported go-live transitions with runbooks, training notes, and performance tuning.
- Managed a team of 3 to deliver an AI-powered video conferencing platform with transcription features; designed backend services and integrations to support real-time workloads and secure access.
- Standardized deployments using containerization (Docker/Kubernetes where applicable) and CI/CD practices to improve release consistency and reduce rollback risk.

Software Engineer | The People Practice (Contract) | Remote (Lagos, Nigeria) | Oct 2023 – Dec 2023

- Designed and implemented 8 APIs to optimize HR Tech solution, improving search functionality and system performance by 35%.

Software Engineer (Full Stack) | ZEDIRAX MULTI CONCEPT | Lagos, Nigeria | Jan 2019 – Dec 2022

- Directed a 15-member development team to deliver a health-tech product on time and within budget.
- Spearheaded end-to-end implementation of SaaS offerings in Python, achieving seamless client integrations and adoption.
- Refactored and developed reusable UI components, improving design consistency and boosting user satisfaction by 70%.
- Identified and remediated a critical security vulnerability, reducing incidents by 50%.
- Architected and maintained websites for Edo State Specialist Hospital and Acaderax, and provided technical support for Ciuci Consulting, enhancing site performance and user engagement.

EDUCATION

Bachelor of Science, Biochemistry | University of Benin, Edo, Nigeria | 2013 – 2017

CERTIFICATIONS

Introducing Generative AI with AWS (UDACITY) | 2025

AWS Educate Machine Learning Foundations (AWS) | 2025

Amazon Elastic Container Service (AWS ECS) | 2023

Google Africa Developer Training Program | 2019

SOME PROJECTS

- [DipoleVibe](#) (2023): Video-sharing and AI meeting recording app. Tech: NodeJS, ReactJS, C#, Python, WebRTC, PostgreSQL, MongoDB, Microservices.
- [Converge ERP](#) (2024): ERP solution managing business processes for Heirs Holdings. Tech: ReactJS, C#, NodeJS, MySQL, Azure VM, Tailwind, ChartJS.
- [Edubanc](#) (2025): Web application offering flexible financing solutions. Tech: NextJS, PostgreSQL, Python, NodeJS.